

Scope Over Monotonicity: NPI Licensing with Czech ‘majority’

Background. Negative Polarity Items (NPIs) are classically analyzed as markers of downward inferences (DIs; Ladusaw 1979), but experimental evidence is mixed. Szabolcsi et al. (2008) found no correlation between DIs and NPI licensing; later studies suggested one based on subjective judgments (Chemla et al. 2011; Denić et al. 2021). Barker (2018) instead argues that NPIs signal *narrow scope* wrt. their licensor, not monotonicity. Dočekal and Chumchalová (2025) likewise found domain-sensitive licensing with upward-entailing (UE) inference patterns. But double licensing is not a *genuinely* non-monotonic (NM) context, so it remains open whether NPIs can appear in *empirically* NM environments (Alexandropoulou et al. 2020). The quantifier *většina* (‘most/majority’) is ideal here: as a *proportional* quantifier (Barwise & Cooper 1981), it is neither upward- nor downward-entailing (DE), with denotation $\llbracket \textit{většina} \rrbracket(A)(B) = 1$ iff $|A \cap B| > |A \setminus B|$. We report a two-part experiment on Czech *sebe-menší* (‘least’) under *většina*. If NPIs are licensed here while subjects reject both DE and UE inferences, this is stronger evidence against the DI–NPI link than DL and supports Barker 2018.

Experiment. The experiment consisted of an acceptability- and an inference judgment task. We hypothesized *sebe-menší* in the restrictor of a *most*-QP would be judged acceptable despite *většina* being seen as NM. Two questionnaire versions were created: *acceptability-first* (N=24) and *inference-first* (N=29). The acceptability part used 9 items in a 1×3 factorial design with conditions: DE *all*, NM *most*, and UE *some*. We expected the descending order of the ratings: ‘all’ > ‘most’ > ‘some’.

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| <p>(1) <i>Všichni/Většina/Někteří občané/-ů se sebe-menší vírou v Bohu chtěli/-a nalezenému dítěti vystrojit křest.</i>
 all/majority/some citizens with <i>sebe</i>-small faith in God wanted found child.DAT arrange christening
 ‘All/Most/Some citizens with even the least faith in God wanted to have the found child christened.’</p> | <p>(2) A: [All/Most/Some] of the circles are in the box on the right.</p> <p>(3) B: ... of the [shapes /green circles] are in the box on the right.</p> |
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The inference task used a 2×3 factorial design: One factor was UE/DE entailment between the presented sentences (*circle* $\rightarrow$ *shape* or *circle* $\rightarrow$ *green.circle*). The other factor was quantifier type (*all/most/some*). Subjects were shown a set of 12 shapes: two types of shapes in red and green. The scenario was that each such set of shapes would be placed in two boxes with opaque walls, with a window in one box that concealed the colors (Fig 1). The task was to answer “yes” or “no” to whether Sentence A entailed Sentence B (see translated example in (2), (3)). We expected mostly “yes” ratings in DE scenarios for *all*, UE for *some*, and barely any “yes” ratings for *most*.

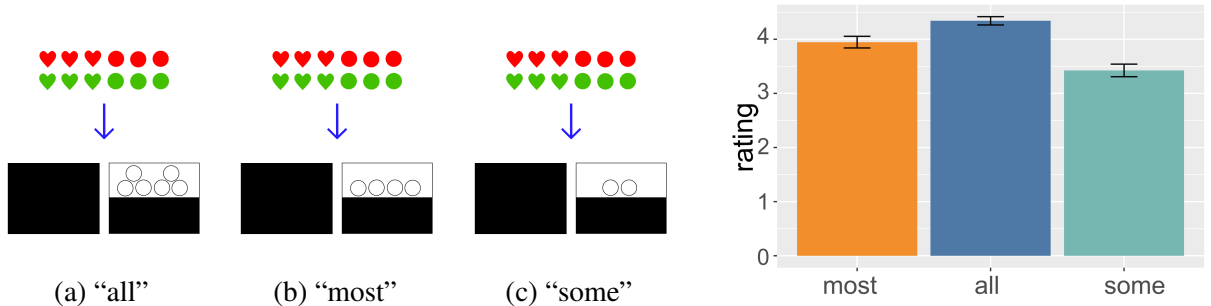


Figure 1: Visuals for the inference task and acceptability ratings with error bars.

Results. Descriptive statistics are shown in Figures 1 and 2. We analyzed acceptability with Bayesian linear mixed models and inference responses with Bayesian logistic mixed models (random intercepts for participants; maximal models did not converge) in *R* using *rstanarm* (Brilleman et al. 2018). Task order showed no credible effect ($|\hat{\beta}| < 0.3$, $\text{BF}_{01} > 4$), so we report

aggregated data. In acceptability, with *most* as the reference level, *all* was rated credibly higher ($\hat{\beta} = 0.43$, 95% CrI [0.18, 0.67], $\text{BF}_{10} = 5.14$) and *some* credibly lower ($\hat{\beta} = -0.49$, 95% CrI [-0.73, -0.24], $\text{BF}_{10} = 24.6$), with extreme evidence for both effects ($\text{pd} > 99.96\%$). For inferences, again with *most* as reference and DE as the directional baseline, participants robustly endorsed DE inferences under *all* ($\hat{\beta}_{\text{UE:all}} = -8.51$, 95% CrI [-11.57, -5.97], $\text{BF}_{10} = 9.20 \times 10^5$) and UE inferences under *some* ($\hat{\beta}_{\text{UE:some}} = 7.09$, 95% CrI [5.35, 9.09], $\text{BF}_{10} = 3.50 \times 10^8$). Critically, *most* showed no evidence for either inference direction (UE: $\hat{\beta} = -1.30$, 95% CrI [-2.85, -0.04], $\text{BF}_{01} = 1.06$), with both DE and UE inferences rejected (DE: 6.8%, UE: 2.3%). Thus, *most* was rated neither DE nor UE despite licensing NPIs at intermediate levels ($\mu=3.95$) between *all* ($\mu=4.34$) and *some* ($\mu=3.42$).

Discussion. The results reveal a critical dissociation: the NM quantifier *most* licenses NPIs at intermediate acceptability levels ($\mu = 3.95$) despite showing no monotonic inference pattern in the same participants (DE: 6.8%; UE: 2.3%). This is at odds with Ladusaw’s (1979) hypothesis that DI are a synchronic licensing condition for NPIs. Unlike DL environments (Homer 2021), *most* is genuinely NM in its restrictor: participants robustly rejected *both* DE and UE inferences ($\text{BF}_{01} = 1.06$), confirming a non-DE/non-UE environment. This dissociation provides stronger evidence against the DI–NPI link than double licensing. Participants showed *extreme* evidence for DE behavior under *all* ($\text{BF}_{10} = 9.20 \times 10^5$) and UE behavior under *some* ($\text{BF}_{10} = 3.5 \times 10^8$), while *most* yielded no credible preference for either direction. The acceptability gradient (*all*: $\mu = 4.34 > \text{most}$: $\mu = 3.95 > \text{some}$: $\mu = 3.42$) mirrors this: maximal DE licensing yields the highest acceptability, genuine NM-ity an intermediate one, and maximal UE the lowest. This pattern is incompatible with Ladusaw’s synchronic DE requirement but aligns with Barker (2018): NPIs signal narrow scope wrt. their licenser (i.e., they are licensed when their wide scope does not entail their narrow scope) rather than monotonicity. Formally, from $\exists d [\text{v} \check{\text{e}}\text{t}\check{\text{s}}\text{i} \text{na}_x (F(x, d) \rightarrow P(x))]$ (“there is a degree of faith such that most people satisfy P ”) one cannot infer $\text{v} \check{\text{e}}\text{t}\check{\text{s}}\text{i} \text{na}_x ((\exists d F(x, d)) \rightarrow P(x))$ (“most people with some degree of faith satisfy P ”)—there may be a threshold of d that guarantees P (first formula true, second false). Under *most*, the NPI takes narrow scope within the restrictor (*sebemenší vírou*, ‘least faith’), yielding an interpretable configuration despite non-monotonicity. The results replicate Szabolcsi et al.’s (2008) null finding and extend it to a theoretically critical case: a quantifier that is *empirically* NM. While NPIs may have grammaticalized in DE environments (Herburger 2023), our data show that DI is not synchronically necessary for licensing.

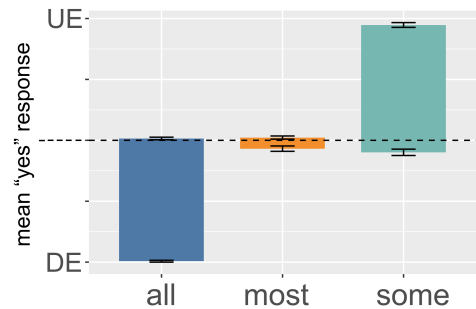


Figure 2: Inference responses.

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